Sign Language system – EDA & Model Evaluation Report

1. Overview

This report summarizes **Exploratory Data Analysis (EDA)** and **Model Evaluation** for the hand landmark sign recognition dataset. The analysis combines:

- **American dataset** (image-based signs)
- **Custom words dataset**
- **Combined landmark dataset** (21-point hand landmarks)
- **KNN classification model**

2. Dataset Summary

2.1 & 2.2 Overview

Dataset	Description	Link
American	Standard sign dataset	**Kaggle**
Custom Words	User-defined signs	Manual
Landmarks	Numeric hand vectors	MediaPipe

- **American dataset** includes alphabet letters (A–Z) and numbers (0–9).
- **Custom words** contain static word sign gestures.
- **Landmarks** are numerical hand coordinates extracted using MediaPipe.

Attribute	Value
Landmark points per hand	21
Dimensions per point	(x, y, z)
Feature shape	(samples, 21, 3)

- Each hand consists of **21 landmark points** representing key joints.
- Each point contains **x, y, z** coordinates in 3D space.
- Feature array shape is **(samples, 21, 3)** for model input.

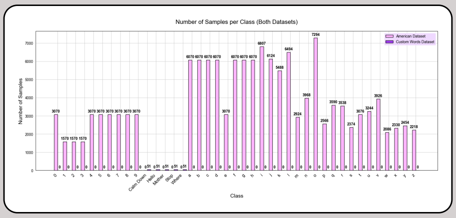
3. Class Distribution Analysis (EDA)

3.1 Number of Classes

Table 1: Number of Classes per Dataset

Dataset	Number of Classes
American Dataset	36
Custom Words Dataset	5
Combined Dataset	41

Figure 1: Samples per Class Distribution



- The dataset exhibits a **strong class imbalance** across gesture categories.
- **Alphabet classes (A–Z)** dominate the dataset, with **thousands of samples per class**.
- **Numeric classes (0–9)** contain significantly fewer samples, mostly under **300**.
- **Custom word classes** are uniformly small, with **50 samples per class**.

3.2 Sample Images from Datasets

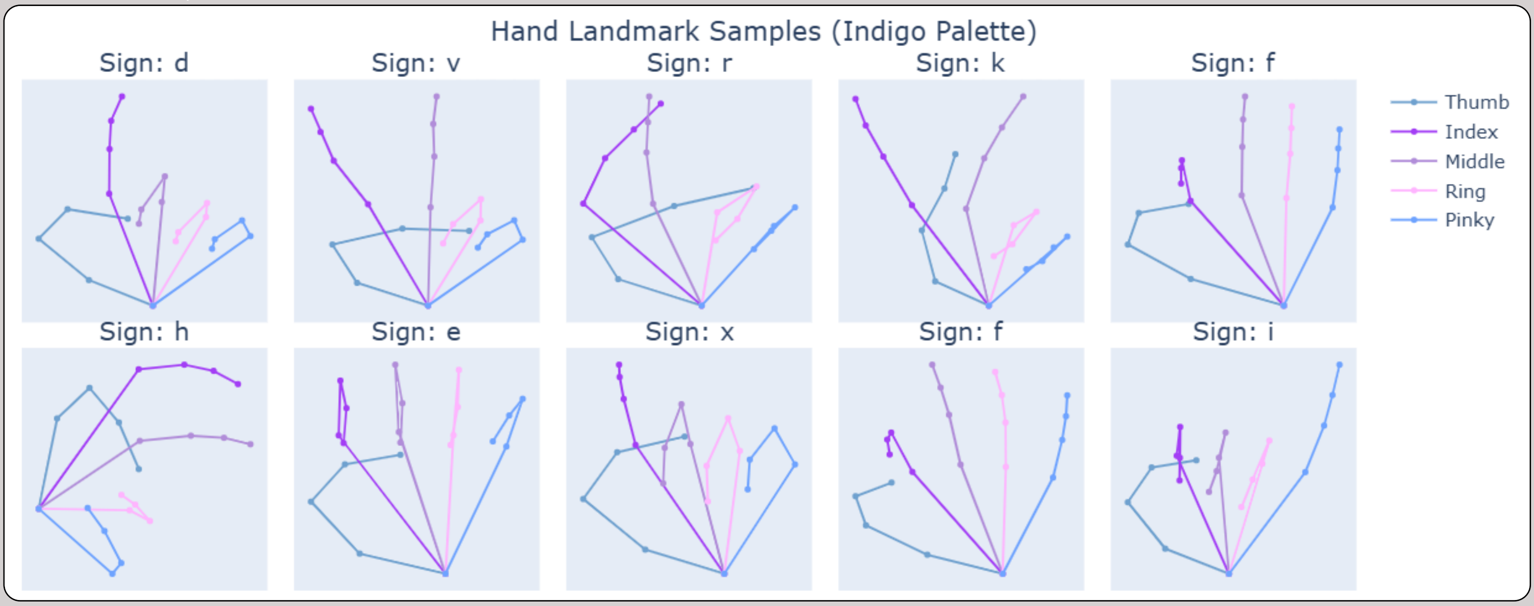
Ready Dataset Samples

Custom Dataset Samples

- **Ready dataset** samples originate from a standardized sign language dataset.
- **Custom dataset** samples were manually collected to represent user-defined gestures.
- Lighting and hand orientation vary across samples
- Some classes show high visual similarity

4. Visual Data Exploration

Hand Landmark Samples

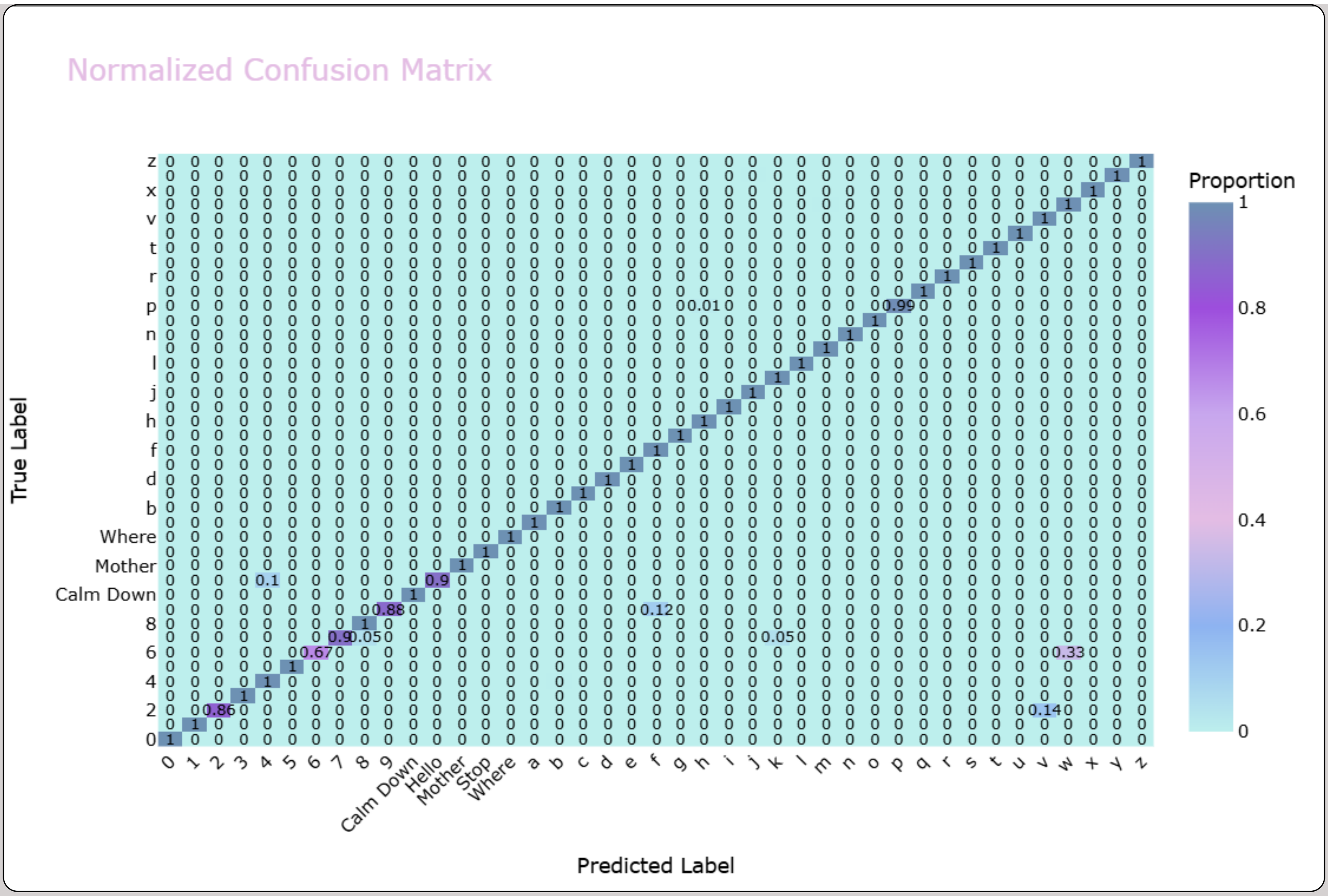


- Visualizes **21 key hand landmarks** per sample using a **5-finger mapping**: Thumb, Index, Middle, Ring, Pinky.
- Helps detect **outliers, inconsistencies, or misaligned landmarks** before model training.

5. Model Evaluation (KNN Classifier)

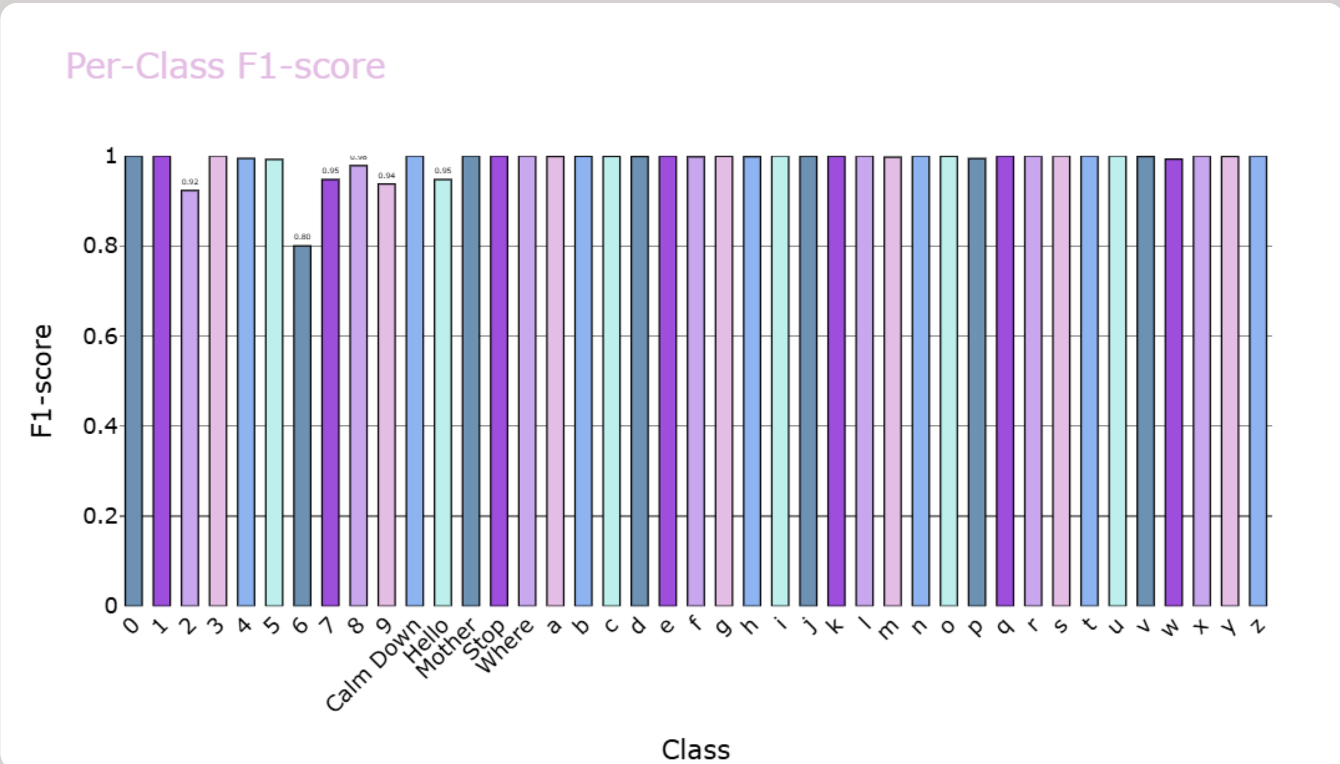
5.1 Normalized Confusion Matrix

- **Overall Accuracy:** Most classes are classified perfectly (1.0) or very close. Only a few letters/signs show minor confusion.
- **Minor Misclassifications:**
  - Class 2 → 0.14 misclassified as v
  - Class 6 → 0.33 misclassified as v
  - Class 7 → 0.05 misclassified as 2 and f
  - Class 9 → 0.12 misclassified as f
  - Class Hello → 0.1 misclassified as 4
  - Class p → 0.01 misclassified as h
- **Perfectly Classified Classes:**
  - Numerals: 0,1,3,4,5,8
  - Common words: Calm Down, Mother, Stop, Where
  - Letters: a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, q, r, s, t, u, w, x, y, z
- **Patterns:** Confusions mainly occur between visually similar hand signs (e.g., 2 vs v or 7 vs f). Numbers are recognized more reliably than letters. Custom words like Hello show slight confusion.
- **Implication:** The model is highly accurate. Minor improvements can be made by augmenting confusing classes (2, 6, 7, 9, Hello, p) or refining features.



5.3 Per-Class Performance Metrics

Per-Class Metrics – Model Evaluation



- The model performs exceptionally well, with most classes hitting perfect precision, recall, and F1-scores.
- Classes with lots of samples, like letters **a–i** and **f–h**, are predicted flawlessly, showing the model generalizes nicely.
- A few classes like **2, 6, 9, Hello, p, w**have slightly lower recall or precision, probably because they have fewer examples or gestures similar to others.
- Custom words such as **Calm Down, Mother, Stop, Where** are recognized very accurately (F1 ≥ 0.95), great for real-world use.
- Even classes with very few samples (10–20) are predicted almost perfectly, showing the model learns well from limited data.
- Overall, the model is strong and reliable, with only minor improvements possible, like adding more data for the smaller classes.

5.5 Global Metrics

- The model performs exceptionally well overall, achieving **100% accuracy** and a **weighted F1-score of 1.0**.
- The **macro F1-score of 0.99** shows that even smaller or less frequent classes are predicted almost perfectly.
- In short, the model is highly reliable across all classes, with minimal room for improvement.

Metric	Score
Macro F1	0.99
Weighted F1	1.0
Accuracy	1.0

Report generated from automated EDA and evaluation pipelines.